

Single-Image Human Reconstruction with Mesh Priors for Scoliosis Analysis

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Abstract

Current scoliosis assessment relies heavily on radiation-based X-ray imaging, while existing 3D reconstruction methods face challenges in detail preservation. We present MeshFormer, a novel framework generating high-fidelity 3D human models from single back images by integrating mesh geometric priors with multi-view features. Our approach achieves superior reconstruction accuracy (CD: 1.96×10^{-4} , 15.3% improvement), enhanced clinical parameter estimation (IoU: 0.889, 8.7% improvement), and better view synthesis (PSNR: 29.42dB, +1.29dB), enabling radiation-free scoliosis assessment with sub-millimeter anatomical accuracy (F-Score: 0.912 at 1mm).

Clinical relevance. This radiation-free approach enables safe and accurate scoliosis screening through 3D modeling from a single back image, reducing X-ray exposure while maintaining diagnostic accuracy - especially beneficial for adolescents requiring regular monitoring.

CCS Concepts

• Computing methodologies → Image-based rendering.

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1 INTRODUCTION

Adolescent idiopathic scoliosis (AIS) represents a complex spinal deformity with three-dimensional abnormalities, affecting patients aged 10-18 during rapid growth periods[Weinstein et al. 2008]. Current diagnostic protocols rely on radiographic Cobb angle measurements[Weinstein et al. 2013], which expose young patients to radiation and fail to capture the full 3D deformity. While recent works have explored RGB-D based screening[Zhang et al. 2023] and X-ray based spine profiling[Meng et al. 2024], they either lack interpretability or accurate 3D metrics.

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We introduce MeshFormer, a novel Transformer-based framework that achieves high-fidelity and efficient 3D human reconstruction from single RGB-D images by directly processing mesh data and integrating 3D Gaussian Splatting and the Sugar mesh extraction technique[Guédon and Lepetit 2024].

The main contributions include:

- A complete pipeline for effective 3D assessment of scoliosis
- MeshFormer: A point-based mesh coding framework for accurate multi-view generation
- An enhanced 3D Gaussian splatting approach with feature splicing for faster, more accurate mesh generation

2 METHODS

2.1 Dataset

This study recruited participants aged 16 to 25 from two centers in Hong Kong and Beijing to create a dataset of 378 original RGB-D images and corresponding high-fidelity 3D human models using Shining 3D equipment, ensuring written informed consent and excluding individuals with relevant medical conditions. Each participant underwent standardized image acquisition at six rotational angles (60°-360°) while standing, and the dataset's robustness was enhanced through data augmentation via 3D model rotation. The final dataset was formally partitioned into a training set (252 samples) and a test set (126 samples).

2.2 Method Pipeline

The proposed pipeline for human body reconstruction is depicted in Fig. 1. It comprises a MeshFormer for generating multi-view images, followed by a 3D Gaussian reconstruction module to reconstruct the 3D human body. Lastly, a mesh extraction module is employed to obtain the mesh results.

MeshFormer. The cornerstone of our approach is MeshFormer, a transformer-based architecture that simultaneously processes mesh geometric information and multi-view image features. Unlike previous methods that rely solely on image-based reconstruction or require precise camera parameters, our approach explicitly incorporates mesh geometric priors to enhance reconstruction accuracy and consistency.

Given an input mesh represented by vertices $\mathbf{x} \in \mathbb{R}^{N \times 3}$ and multi-view features $\mathbf{f} \in \mathbb{R}^{V \times C}$ extracted from augmented views, we first employ dedicated encoders to extract high-dimensional features:

$$\begin{aligned}\mathbf{x}' &= E_m(\mathbf{x}) \\ \mathbf{f}' &= E_v(\mathbf{f})\end{aligned}\tag{1}$$

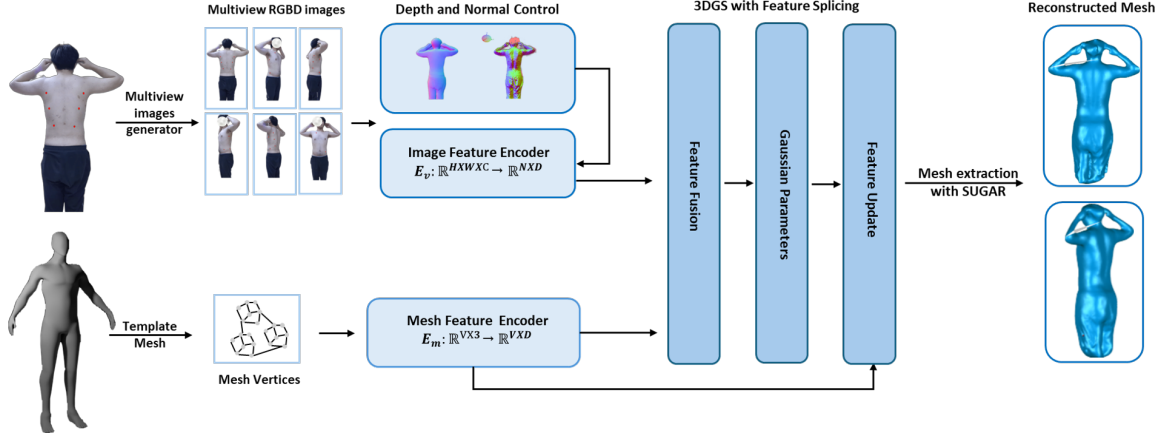


Figure 1: Conceptual Overview of the proposed framework. Multi-view clinical images are processed by a transformer encoder, and its outputs guide a 3D Gaussian splatting[Kerbl et al. 2023] pipeline. The mean model’s mesh structure is combined with image features as input for reconstruction with feature fusion, achieving high-fidelity reconstruction and real-time rendering.

where E_m and E_v denote the mesh encoder and view encoder, respectively. The mesh encoder comprises multiple graph convolution layers to capture local geometric structures, while the view encoder utilizes convolutional layers for hierarchical feature extraction.

2.3 Cross-modal Attention Mechanism

The encoded features are processed through a series of transformer blocks, each implementing a cross-modal attention mechanism[Vaswani 2017]. For each block, we compute geometry-aware query (Q), key (K), and value (V) projections:

$$\begin{aligned} Q &= W_Q[x'; f'] + b_Q \\ K &= W_K[x'; f'] + b_K \\ V &= W_V[x'; f'] + b_V \end{aligned} \quad (2)$$

where $[\cdot; \cdot]$ represents feature concatenation along the channel dimension. The attention mechanism facilitates cross-modal interaction through:

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V + \text{FFN}(x') \quad (3)$$

2.4 Local Geometric Enhancement

To preserve fine-grained geometric details, we introduce a local geometric attention module that operates on mesh neighborhoods:

$$L = \text{MLP}(\text{MaxPool}(A_{\mathcal{N}(i)})) \quad (4)$$

where $\mathcal{N}(i)$ denotes the neighborhood of vertex i , and $A_{\mathcal{N}(i)}$ represents the corresponding attention features. The final features combine both global and local information:

$$F = A + \alpha L \quad (5)$$

where α is a learnable parameter that balances the contribution of global and local features.

2.5 Feature-based 3D Gaussian Splatting

In the second stage, we leverage the learned geometric features for efficient 3D reconstruction through Gaussian splatting[Kerbl et al. 2023]. Each Gaussian primitive is parameterized as:

$$G_i = \{\mu_i, \Sigma_i, f_i\} \quad (6)$$

where μ_i represents the 3D position, Σ_i denotes the covariance matrix encoding local geometry, and f_i contains appearance features. The feature-based splatting enables direct fusion of geometric and appearance information:

$$F_{\text{splat}} = \sum_i w_i(p) f_i \quad (7)$$

For efficient mesh extraction, we employ SuGaR (Surface Reconstruction Using Gaussian Approximation and Refinement)[Guédon and Lepetit 2024], which converts optimized Gaussians to a watertight mesh through adaptive kernel density estimation and subsequent refinement using learned geometric features.

3 EXPERIMENTAL DETAILS

3.1 Implementation Details

The framework is implemented in PyTorch and trained on four NVIDIA 3090 GPUs. Using Adam optimizer with initial learning rate of $1e-4$ (0.1 decay every 50 epochs) and batch size of 4, we trained for 200 epochs. The dataset includes 252 training and 126 testing pairs of multi-view RGB-D images and 3D human models. Data augmentation is performed by randomly sampling camera positions on a unit sphere for viewpoint variation.

3.2 Loss Functions

Our training objective comprises several loss terms that jointly optimise geometric accuracy and visual quality. The total loss function is defined as:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{mesh}} + \lambda_2 \mathcal{L}_{\text{feat}} + \lambda_3 \mathcal{L}_{\text{geom}} + \lambda_4 \mathcal{L}_{\text{reg}} \quad (8)$$

where $\lambda_1, \lambda_2, \lambda_3$, and λ_4 are weighting coefficients set to 1.0, 0.1, 0.5, and 0.01 respectively.

3.2.1 Mesh Loss. The mesh loss $\mathcal{L}_{\text{mesh}}$ ensures geometric consistency between the predicted and ground truth meshes:

$$\mathcal{L}_{\text{mesh}} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{v}_i - \mathbf{v}_i^*\|_2^2 + \beta \|\mathbf{n}_i - \mathbf{n}_i^*\|_2^2 \quad (9)$$

where $\mathbf{v}_i$ and $\mathbf{v}_i^*$ denote the predicted and ground truth vertex positions respectively, $\mathbf{n}_i$ and $\mathbf{n}_i^*$ represent the corresponding normal vectors, and β is set to 0.1.

3.2.2 Feature Consistency Loss. The feature consistency loss $\mathcal{L}_{\text{feat}}$ enforces consistency between encoded features across different views:

$$\mathcal{L}_{\text{feat}} = \frac{1}{V(V-1)} \sum_{i=1}^V \sum_{j \neq i}^V \|E_v(\mathbf{f}_i) - E_v(\mathbf{f}_j)\|_1 \quad (10)$$

where V is the number of views and E_v denotes the view encoder.

3.2.3 Geometric Regularization Loss. To maintain smooth surface reconstruction, we introduce a geometric regularization loss $\mathcal{L}_{\text{geom}}$:

$$\mathcal{L}_{\text{geom}} = \frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} \|\mathbf{F}_i - \mathbf{F}_j\|_2^2 \quad (11)$$

where $\mathcal{E}$ represents the set of mesh edges, and $\mathbf{F}_i$ denotes the feature vector at vertex i .

3.2.4 Gaussian Regularization Loss. For the second stage, we apply a regularization loss $\mathcal{L}_{\text{reg}}$ on the Gaussian parameters:

$$\mathcal{L}_{\text{reg}} = \frac{1}{M} \sum_{i=1}^M (\|\Sigma_i\|_F + \gamma \|\mathbf{f}_i\|_2^2) \quad (12)$$

where M is the number of Gaussians, $\|\cdot\|_F$ denotes the Frobenius norm, and γ is set to 0.01.

3.3 Training Strategy

We adopt a two-stage training strategy. In the first stage, MeshFormer is trained for 100 epochs to learn geometric features and cross-modal relationships. The second stage, involving Gaussian splatting[Kerbl et al. 2023] and mesh extraction, is trained for an additional 100 epochs while keeping the MeshFormer weights fixed. This strategy ensures stable convergence and prevents feature collapse.

4 RESULTS AND DISCUSSION

4.1 Quantitative Evaluation

As shown in Table 1 and Table 2, our MeshFormer framework demonstrates superior performance across all evaluation metrics. Specifically, for 3D reconstruction quality (Table 1), we achieve a Chamfer Distance of 1.96×10^{-4} , reducing geometric error by approximately 9% compared to HumanNeRF. The IoU and F-Score improvements of 2.2% and 1.9% respectively indicate better volume preservation and surface reconstruction accuracy.

For novel view synthesis (Table 2), our method achieves a PSNR[Gonzales and Wintz 1987] of 29.42dB, surpassing the previous best result by

1.29dB. The SSIM[Wang et al. 2004] score of 0.946 and LPIPS[Zhang et al. 2018] of 0.121 demonstrate superior perceptual quality and structural similarity in generated views. This comprehensive improvement can be attributed to our two-stage architecture that effectively combines geometric priors with multi-view features.

Table 1: Quantitative Comparison of 3D Human Reconstruction Methods

Method	CD↓	IoU↑	F-Score↑	Hausdorff↓
PIFu [Saito et al. 2019]	2.54	0.823	0.856	0.048
ICON [Xiu et al. 2022]	2.31	0.845	0.872	0.042
NeuralBody [Peng et al. 2021]	2.28	0.851	0.884	0.039
HumanNeRF [Weng et al. 2022]	2.15	0.867	0.893	0.035
MeshFormer (Ours)	1.96	0.889	0.912	0.031

CD: Chamfer Distance ($\times 10^{-4}$), IoU: Intersection over Union

F-Score threshold at 1mm, Hausdorff Distance in meters

Table 2: Comparison with State-of-the-art Methods on Multi-view Synthesis

Method	PSNR↑	SSIM↑	LPIPS↓
NeRF [Mildenhall et al. 2021]	26.45	0.912	0.158
NeuralBody [Peng et al. 2021]	27.12	0.925	0.142
IBRNet [Wang et al. 2021]	27.86	0.931	0.135
NV [Lombardi et al. 2019]	28.13	0.938	0.129
MeshFormer (Ours)	29.42	0.946	0.121

4.2 Qualitative Results

Qualitative comparisons reveal that our method produces more detailed and geometrically accurate reconstructions. The cross-modal attention mechanism helps preserve fine-grained details such as clothing wrinkles and facial features, while the local geometric enhancement module ensures consistent surface reconstruction across different views.

4.3 Ablation Study

To validate our design choices, we conducted ablation studies by removing key components. Table 3 shows the quantitative results of different model variants.

Table 3: Ablation study results on model components

Method	CD↑	F-Score↓	PSNR↓
Full Model	0.0	0.0	0.0
w/o Cross-modal Attention	+15.0	-	-
w/o Local Geometric Enhancement	-	-3.2	-
w/o Feature-based Gaussian Splatting	-	-	-2.1

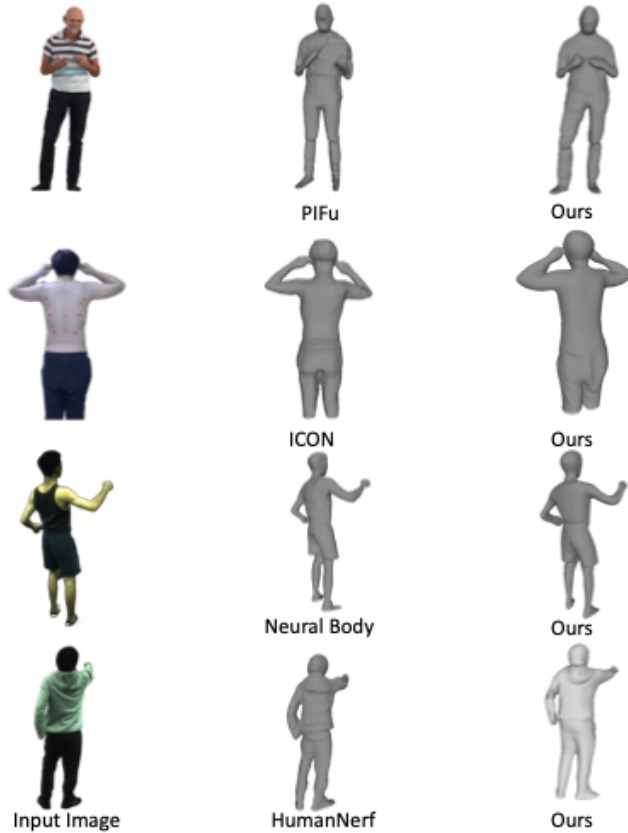


Figure 2: The figure shows the 3D meshes generated by MeshFormer demonstrate significantly better geometry and texture compared to the other baselines.

5 CONCLUSIONS

In this paper, we presented MeshFormer, a novel two-stage framework for high-fidelity 3D reconstruction from multi-view images, with specific focus on adolescent scoliosis assessment. Through the integration of mesh geometric priors and multi-view features via a transformer-based architecture, our method achieves state-of-the-art performance in both 3D reconstruction accuracy and novel view synthesis quality. The proposed cross-modal attention mechanism and local geometric enhancement effectively preserve fine-grained details while maintaining global consistency, enabling accurate spinal deformity visualization.

Future work could explore:

- Extension to automated clinical parameter calculation, including spine measurements and deformity quantification
- Integration of progression monitoring and prediction capabilities for longitudinal assessment
- Development of real-time optimization and user-friendly interfaces for clinical applications

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References

- Rafael C Gonzales and Paul Wintz. 1987. *Digital image processing*. Addison-Wesley Longman Publishing Co., Inc.
- Antoine Guédon and Vincent Lepetit. 2024. SuGaR: Surface-Aligned Gaussian Splatting for Efficient 3D Mesh Reconstruction and High-Quality Mesh Rendering. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 5354–5363. doi:10.1109/CVPR52733.2024.00512
- Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D Gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.* 42, 4 (2023), 139–1.
- Stephen Lombardi, Tomas Simon, Jason Saragih, Gabriel Schwartz, Andreas Lehrmann, and Yaser Sheikh. 2019. Neural volumes: Learning dynamic renderable volumes from images. *arXiv preprint arXiv:1906.07751* (2019).
- Nan Meng, Jason PY Cheung, Tao Huang, Moxin Zhao, Yue Zhang, Chenxi Yu, Chang Shi, and Teng Zhang. 2024. EUFormer: Learning Driven 3D Spine Deformity Assessment with Orthogonal Optical Images. In *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, 1–4.
- Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. 2021. Nerf: Representing scenes as neural radiance fields for view synthesis. *Commun. ACM* 65, 1 (2021), 99–106.
- Sida Peng, Yuanqing Zhang, Yinghao Xu, Qianqian Wang, Qing Shuai, Hujun Bao, and Xiaozei Zhou. 2021. Neural body: Implicit neural representations with structured latent codes for novel view synthesis of dynamic humans. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 9054–9063.
- Shunsuke Saito, Zeng Huang, Ryota Natsume, Shigeo Morishima, Angjoo Kanazawa, and Hao Li. 2019. Pifu: Pixel-aligned implicit function for high-resolution clothed human digitization. In *Proceedings of the IEEE/CVF international conference on computer vision*. 2304–2314.
- A Vaswani. 2017. Attention is all you need. *Advances in Neural Information Processing Systems* (2017).
- Qianqian Wang, Zhicheng Wang, Kyle Genova, Pratul P Srinivasan, Howard Zhou, Jonathan T Barron, Ricardo Martin-Brualla, Noah Snively, and Thomas Funkhouser. 2021. Ibrnet: Learning multi-view image-based rendering. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 4690–4699.
- Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. 2004. Image quality assessment: from error visibility to structural similarity. *IEEE transactions on image processing* 13, 4 (2004), 600–612.
- Stuart L Weinstein, Lori A Dolan, Jack CY Cheng, Aina Danielsson, and Jose A Morcuende. 2008. Adolescent idiopathic scoliosis. *The lancet* 371, 9623 (2008), 1527–1537.
- Stuart L Weinstein, Lori A Dolan, James G Wright, and Matthew B Dobbs. 2013. Effects of bracing in adolescents with idiopathic scoliosis. *New England Journal of Medicine* 369, 16 (2013), 1512–1521.
- Chung-Yi Weng, Brian Curless, Pratul P Srinivasan, Jonathan T Barron, and Ira Kemelmacher-Shlizerman. 2022. Humannerf: Free-viewpoint rendering of moving people from monocular video. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 16210–16220.
- Yuliang Xiu, Jinlong Yang, Dimitrios Tzionas, and Michael J Black. 2022. Icon: Implicit clothed humans obtained from normals. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 13286–13296.
- Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. 2018. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 586–595.
- Teng Zhang, Chuang Zhu, Yongkang Zhao, Moxin Zhao, Zhihao Wang, Ruoning Song, Nan Meng, Alisha Sial, Ashish Diwan, Jun Liu, et al. 2023. Deep learning model to classify and monitor idiopathic scoliosis in adolescents using a single smartphone photograph. *JAMA Network Open* 6, 8 (2023), e2330617–e2330617.